Fair for Whom?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Use fractions: 3 successes out of 5 is 3/5, not 3/2.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Compute a correct-out-of-assessed fraction and explain its denominator. Try the recall before opening notes.

Your learning target

I can compute a correct-out-of-assessed fraction and explain its **denominator**.

I can compare systems on shared cases while checking performance within conditions.

I can combine counts correctly and keep missing outcomes separate from assessed errors.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

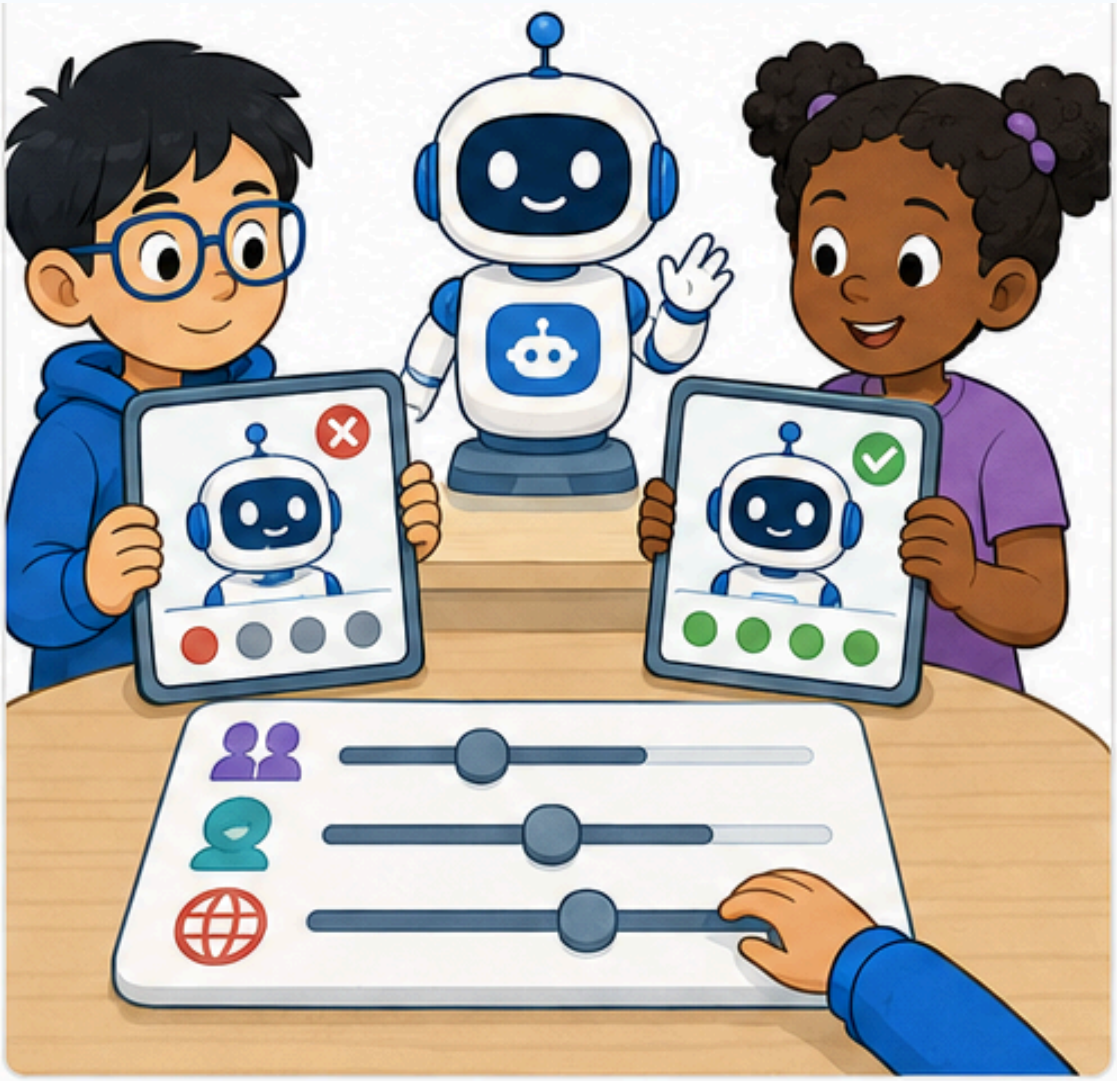
Without opening the old lesson, explain one key idea from Missing and Unbalanced Examples. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Prediction** Is Not Fact. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Fairness requires checking outcomes for different users and situations.



Your target

LEARNING TARGET System A succeeds on 6 of 8 quiet tests and 3 of 4 noisy tests. Is its noisy success rate lower? Show fractions and percentages. What can these small tests not establish?

Learner name:

Readiness for the connected explanation

Eight cases were selected, but one has no assessed outcome. How many have assessed outcomes, and what must a performance report say about the remaining case?

This week’s end goal

By the end, I can compare group fractions and combine unequal-sized groups correctly.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Fair for Whom? case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY A voice game may work well overall yet miss more commands from quiet speakers than loud speakers.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

For this week, **accuracy** means the number of correctly labelled cases divided by the number of assessed cases.

Every assessed case has a system label and a checked reference label, so its result can be judged correct or incorrect. State the system version, cases and scoring rule with the fraction.

Cases without an assessable outcome are reported separately.

Two fixed shape-recognition versions, A and B, label the same 12 previously unseen picture cards. Eight cards are bright and four are dim; each has a checked circle or square reference label.

Both versions return labels for all 12. A is correct on 8 bright and 1 dim card. B is correct on 6 bright and 2 dim cards.

A therefore has 9 correct out of 12 assessed, written 9/12. Dividing **numerator** and **denominator** by 3 gives 3/4. Three quarters equals 75 out of 100, or 75%. B has 8/12, equal to 2/3.

For this comparison keep B as an exact fraction; a rounded percentage is unnecessary.

STUDENT LESSON · 3 OF 18

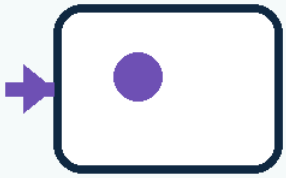
See how it works

> Follow the mechanism for Fair for Whom?. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



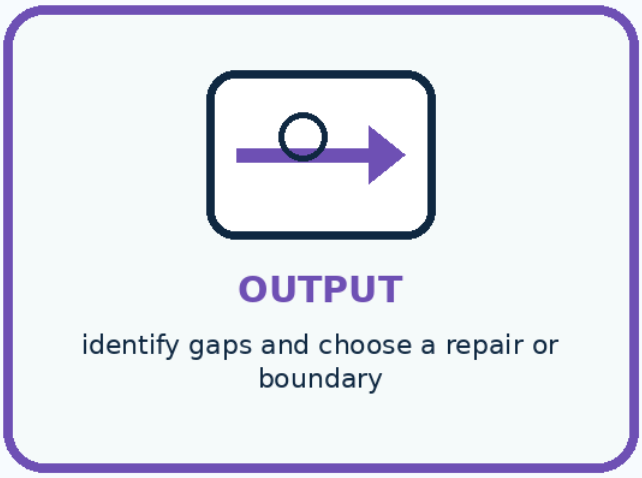
INPUT

test cases grouped by relevant user or condition



PROCESS

compare outcomes and error types by group



INPUT test cases grouped by relevant user or condition

PROCESS compare outcomes and error types by group

OUTPUT identify gaps and choose a repair or boundary

Key idea

CORE IDEA **Fairness** requires checking outcomes for different users and situations.

How the mechanism works

UNDER THE HOOD **Fairness** is context-specific and cannot be established by overall **accuracy** alone. Compare performance, **error cost**s, access, and remedy across relevant groups while avoiding stereotypes and protecting privacy.

Draw a fresh example with all three stages:

A closer explanation

Combine correct and assessed counts

CONDITION	VERSION A	VERSION B
Bright: 8 cards	8 correct of 8	6 correct of 8
Dim: 4 cards	1 correct of 4	2 correct of 4
All: 12 cards	9 of 12 = 3 of 4	8 of 12 = 2 of 3

A has the higher overall accuracy on these same 12 cards, while B has the higher accuracy on the 4 dim cards. The overall fraction does not describe every condition equally.

With the same **denominator** 12, compare the **numerator**s: 9 is greater than 8. A is correct on one more of these cards.

This establishes the ordering on this test under this scoring rule; it does not prove that A will always outperform B on future cards.

To compare fractions with different **denominator**s, rewrite them as **equivalent fractions** with a common **denominator**. For example, 3/4 equals 9/12 and 2/3 equals 8/12.

Changing the written **denominator** does not create new test cases: it only helps compare the proportions.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

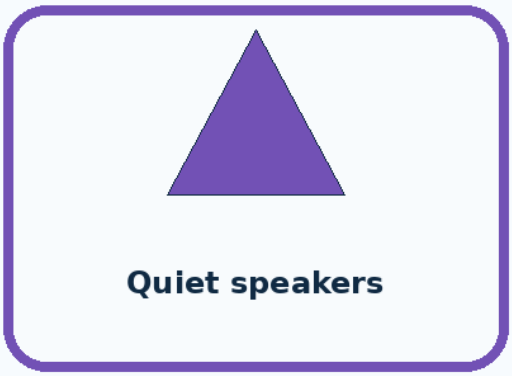
> Compare the supplied cases for Fair for Whom?. Identify the detail that changes the conclusion.

PICTURE GALLERY

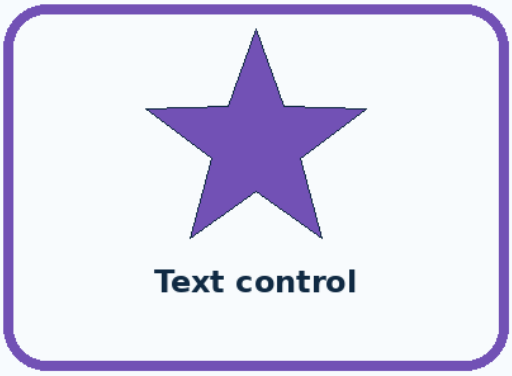
Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Loud speakers Clue: 9 of 10 commands work. Meaning: high success.

The case to investigate

CASE 2 — Quiet speakers Clue: 5 of 10 commands work. Meaning: performance gap.

The case to investigate

CASE 3 — Overall 14 of 20 Clue: 70 percent total. Meaning: hides difference.

The case to investigate

CASE 4 — Text control Clue: alternate access path. Meaning: possible remedy.

Keep this boundary clear

BOUNDARY **Fairness** work should reduce burdens without turning group labels into stereotypes.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Give bright and dim groups equal weight as an explicitly different metric.

Predicted result

Prediction: M scores **70%**; N scores about 81.7%.

Why the result follows

Reason: This averages group rates and is not the original 16-card agreement rate.

Words that unlock the idea

> Use the vocabulary of Fair for Whom? to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
fairness	reasoned treatment of outcomes across people and situations	
subgroup result	performance measured for one relevant group	
gap	difference between group outcomes	
error cost	harm or burden caused by a mistake	

Misconception detective

Find the unsupported claim

CLAIM TO REPAIR Good overall **accuracy** proves **fairness**.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Numerator	The count above the fraction line; here, assessed correct cases
Denominator	The count below the fraction line; here, all assessed cases
Equivalent fractions	Different written fractions that represent the same proportion
Accuracy	The fraction of assessed classification cases labelled correctly under a stated rule
Aggregate	A result formed by combining the counts of several groups

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STUDENT LESSON · 6 OF 18

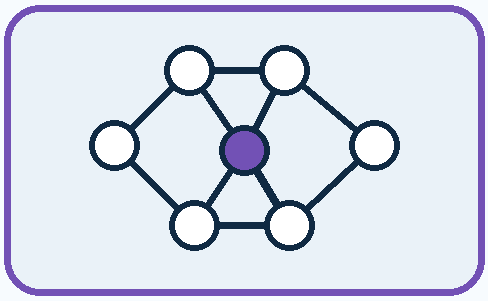
Follow a worked example

> Follow the worked trace for Fair for Whom?, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE A voice game may work well overall yet miss more commands from quiet speakers than loud speakers.

Step 1

STEP 1 — OBSERVE test cases grouped by relevant user or condition

Step 2

STEP 2 — TRACE compare outcomes and error types by group

Step 3

STEP 3 — CONCLUDE identify gaps and choose a repair or boundary

Model answer

MODEL REASONING Report each group's success count and error pattern, not only the total.

Improve microphone/**process** testing with consented varied voices and keep a non-voice control so access does not depend on one **input** mode.

A claim to avoid

DO NOT OVERCLAIM **Fairness** work should reduce burdens without turning group labels into stereotypes.

What new **evidence** could change this answer?

A closer explanation

Question	Version A	Version B
Bright-card accuracy	8/8 = 1	6/8 = 3/4
Dim-card accuracy	1/4	2/4 = 1/2
Combined accuracy	(8 + 1)/(8 + 4) = 9/12	(6 + 2)/(8 + 4) = 8/12

Worked explanation · Fair for Whom?

The bright group has twice as many assessed cards as the dim group. Each individual card counts once in the combined result, so add correct counts and assessed counts.

Do not simply average the two group percentages and call it the all-card **accuracy**: that would give each group equal weight despite their different sizes.

Worked explanation · Fair for Whom?

If the intended task especially needs dim-card performance, inspect that result and gather suitable additional **evidence** before choosing. An overall leader can still be weaker in a relevant condition.

This table alone also does not tell us whether particular mistakes have different consequences. The scoring rule must match the purpose.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Fair for Whom?

A fictional badge reader is evaluated on bright and dim cards with known reference labels.

The facts we will use

Supplied facts

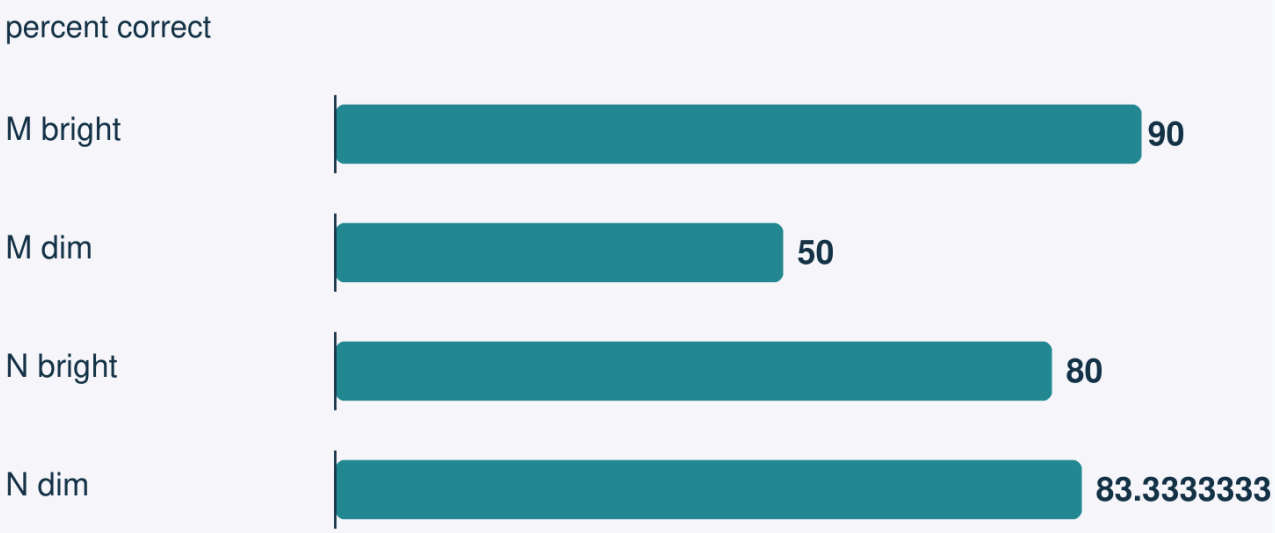
Version M is correct on 9 of 10 bright cards and 3 of 6 dim cards.

Supplied facts

Version N uses the same cards: 8 of 10 bright and 5 of 6 dim correct.

AI ACADEMY / AI-1/23

Group rates compared on the same cards



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/23

Fair for Whom?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Bright group

M:9/10=90%; N:8/10=80%.



2. Dim group

M:3/6=50%; N:5/6≈83.3%.



3. Combined counts

M:(9+3)/16=75%; N:(8+5)/16=81.25%.



4. Discuss trade-off

N improves total and dim results but loses one bright success.

CHANGE ONE THING

Give bright and dim groups equal weight as an explicitly different metric.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Bright group	M:9/10=90%; N:8/10=80%.
2. Dim group	M:3/6=50%; N:5/6≈83.3%.
3. Combined counts	M:(9+3)/16=75%; N:(8+5)/16=81.25%.
4. Discuss trade-off	N improves total and dim results but loses one bright success.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Adding correct counts and **denominator**s weights each tested card once. Averaging 90% and 50% would incorrectly give unequal groups equal size in the all-card rate. The group breakdown reveals a trade-off hidden by the total.

Decide what matters with the relevant users rather than announcing that one number proves **fairness**.

What this example does not establish

Keep the stated limit

Six dim cards supply limited **evidence**; these descriptive rates do not prove a system is fair for all users.

Keep the stated limit

Read the labelled picture: Which group loses performance under N, and can total improvement erase that fact?

Keep the stated limit

Change one thing: Give bright and dim groups equal weight as an explicitly different metric. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Fair for Whom? claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Loud speakers	9 of 10 commands work	high success
Quiet speakers	5 of 10 commands work	performance gap
Overall 14 of 20	70 percent total	hides difference
Text control	alternate access path	possible remedy

Repair two tempting claims

Challenge claim A

CLAIM A Good overall **accuracy** proves **fairness**.

Repair:

Challenge claim B

CLAIM B **Fairness** means every group must have identical data or outcomes in every situation.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

A useful performance statement gives the correct count, assessed count, missing count and evaluation conditions. For A: 9/12 correct, no missing outcomes, on the shared test of 8 bright and 4 dim cards.

Follow it with the important contrast: 8/8 bright and 1/4 dim. Percentages are optional shorthand, not replacements for these counts.

Check the claim against the facts

A score of 3/4 and a score of 75/100 are equal proportions, but they describe different quantities of **evidence** if they come from different tests. The collection method and case coverage still matter.

Neither fraction is an individual **prediction** confidence or a guarantee about the next case.

A system has 2 correct of 2 in one group and 4 correct of 8 in another. Write the combined correct and assessed counts. Explain why averaging 100% and 50% would answer a different question

STUDENT LESSON · 8 OF 18

Practise together

Practise Fair for Whom? with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: **Fairness** requires checking outcomes for different users and situations.

Use a fresh example to explain **fairness**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: **Fairness** means every group must have identical data or outcomes in every situation.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

A third version C is run on the same 12 cards, but two results are lost before they can be judged. Among its 10 assessed outcomes, 9 are correct and 1 is wrong: 9/10 = 90% assessed **accuracy**, with 2 outcomes unassessed.

Do not call this 90% on all 12. Its assessed subset differs from A and B, so the headline fractions do not give the same complete-case comparison.

> **A report declares C best because 90% exceeds A's 75%. Write the fraction and missing count for C, explain the comparison problem, and propose an evaluation step that would make the evidence easier to compare**

Recover and assess the two lost results if possible, without changing the versions.

Another clearly labelled comparison can use the exact same cases with known outcomes for every version; disclose which cases it excludes and why. The current totals do not tell us how A performed on C's ten assessed cases.

Never invent that subset score.

Practise with less help: Compare totals and conditions

Use workbook W2 for the connected example "Fractions and performance comparisons". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Fractions and performance comparisons. Proposed explanation: "Only correct cases belong in the **denominator**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Which group loses performance under N, and can total improvement erase that fact?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

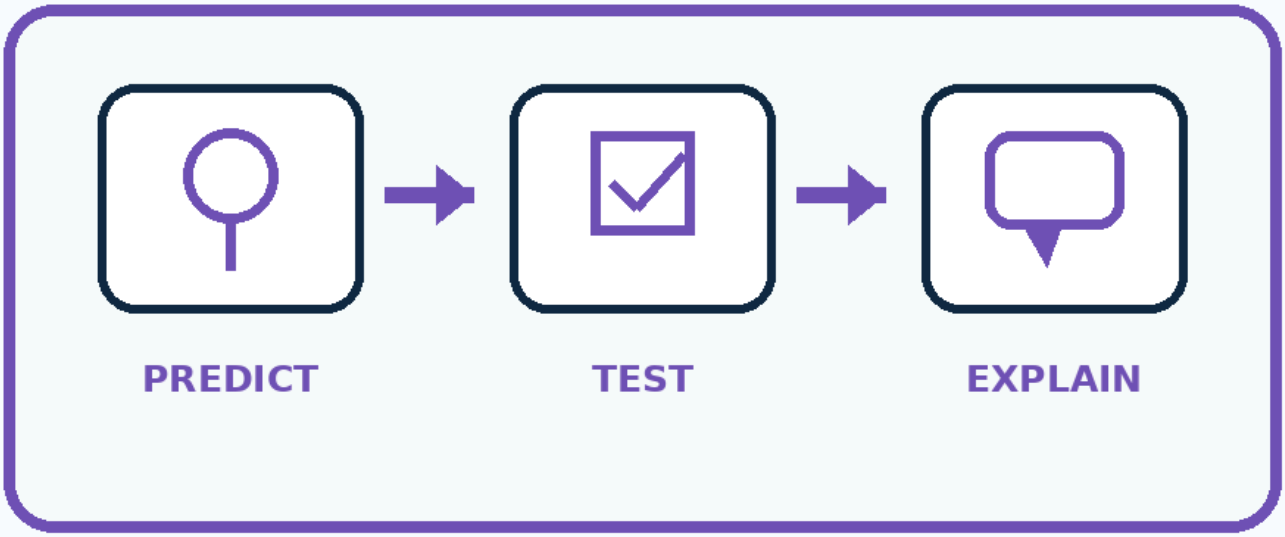
Run the investigation

> Investigate Fair for Whom?. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Voice Game Fairness Check

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Run a voice-game test for quiet and loud speakers.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

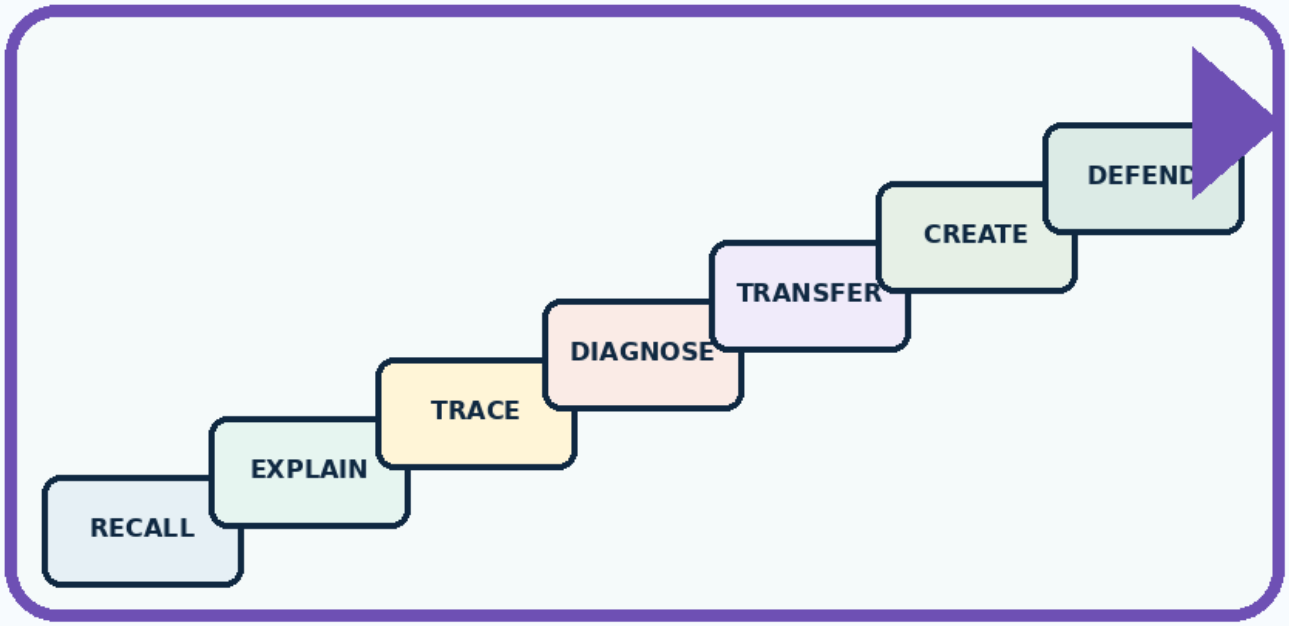
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Fair for Whom?. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **fairness**.

EXPLAIN **Fairness** requires checking outcomes for different users and situations.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: Good overall **accuracy** proves **fairness**.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

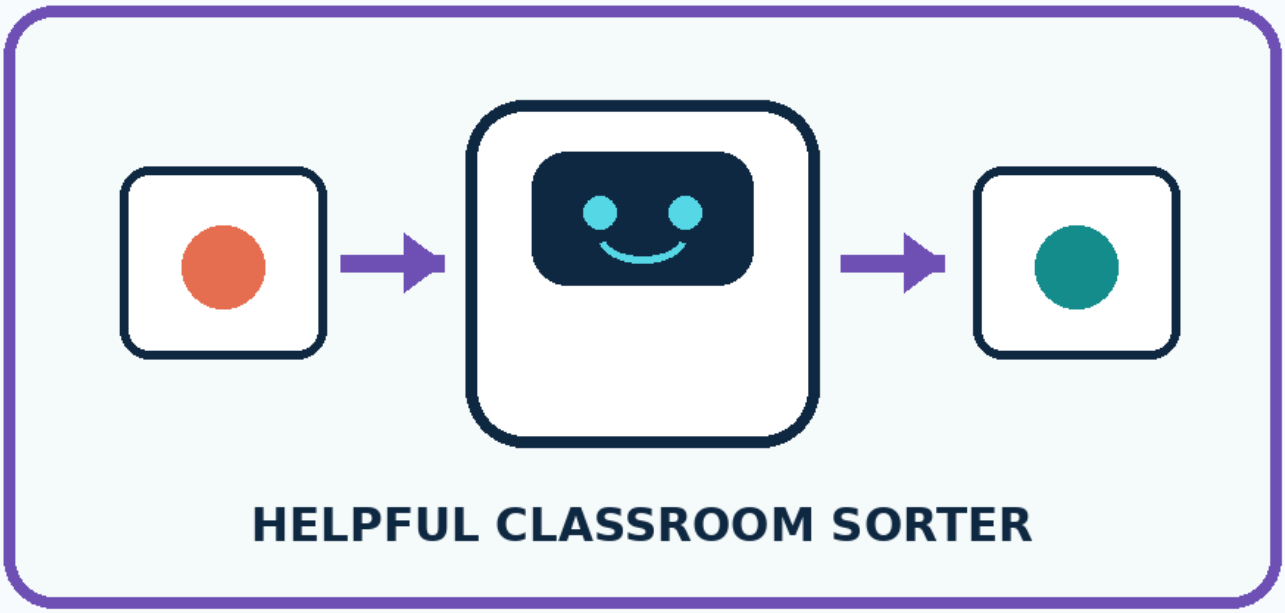
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Test the sorter across relevant item conditions and users; never infer sensitive identity to create a **fairness** check.

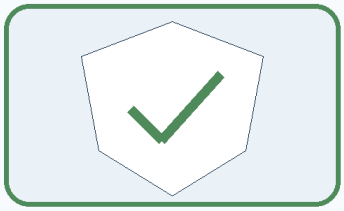
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN **Fairness** work should reduce burdens without turning group labels into stereotypes.

My project decision:

Evidence for your project

Report sorter outcomes by relevant conditions and record whose failures matter when choosing a revision.

STUDENT LESSON · 12 OF 18

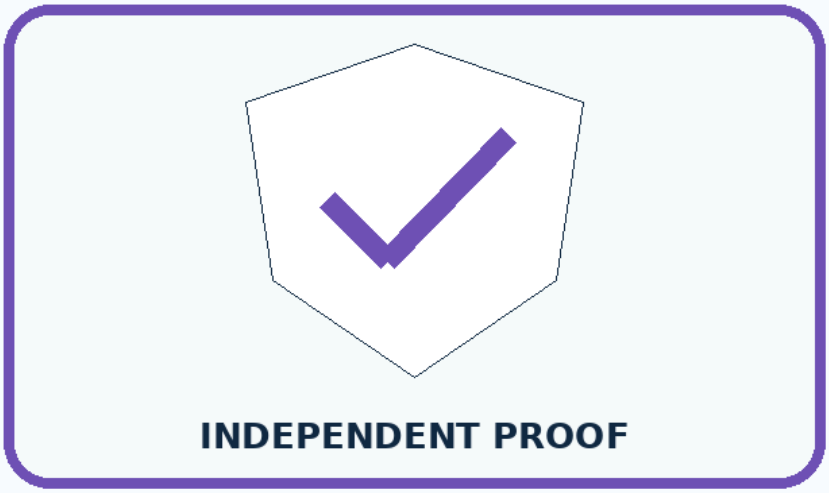
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Test quiet and loud speaker cases separately, compare results, and propose one repair plus one alternate access path.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or **fairness**/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK Report each group's success count and error pattern, not only the total. Improve microphone/**process** testing with consented varied voices and keep a non-voice control so access does not depend on one **input** mode.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Privacy by Choosing Less.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Fair for Whom? case for supported practice. Repeated examples do not become fresh assessments.

Week 23 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A voice system succeeds on 8/10 quiet and 4/10 noisy recordings.

The question

Compare and propose the next investigation.

Worked explanation

The rates differ. Test more varied recordings under controlled conditions and inspect other factors before claiming a cause.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Compare success under two conditions

Evidence or stage	Value or result
Quiet	8/10
Noisy	4/10

A voice system succeeds on 8/10 quiet and 4/10 noisy recordings.

Compare and propose the next investigation.

The rates differ. Test more varied recordings under controlled conditions and inspect other factors before claiming a cause.

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Compare a new shared sound test

Versions X and Y label the same 20 previously unseen clips: 12 quiet and 8 noisy, with no missing outcomes. X is correct on 10 quiet and 2 noisy clips; Y on 9 quiet and 5 noisy clips.

Calculate each combined fraction and percentage, identify the overall leader and compare the leaders within each group. Explain why an overall result cannot guarantee the next clip's outcome.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 23 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A helper succeeds on 4 of 4 clear voices and 1 of 4 quiet voices. Who is less well served? What **evidence** should a repair preserve?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Test the sorter across relevant item conditions and users; never infer sensitive identity to create a **fairness** check.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

> System A succeeds on 6 of 8 quiet tests and 3 of 4 noisy tests. Is its noisy success rate lower? Show fractions and percentages. What can these small tests not establish?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Fair for Whom?, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional badge reader is evaluated on bright and dim cards with known reference labels.

By the end, I can	How I will show it
Compute a correct-out-of-assessed fraction and explain its denominator	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Compare systems on shared cases while checking performance within conditions	Explain the deciding step, show a trace or test, and state one limit.
Combine counts correctly and keep missing outcomes separate from assessed errors	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Fair for Whom? without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Compute a correct-out-of-assessed fraction and explain its **denominator**

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional badge reader is evaluated on bright and dim cards with known reference labels.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? N improves total and dim results but loses one bright success.

1. Dim group

2. Combined counts

3. Bright group

4. Discuss trade-off

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Discuss trade-off N improves total and dim results but loses one bright success. Adding correct counts and **denominator**s weights each tested card once.

Averaging 90% and 50% would incorrectly give unequal groups equal size in the all-card rate. The group breakdown reveals a trade-off hidden by the total.

Decide what matters with the relevant users rather than announcing that one number proves **fairness**.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Give bright and dim groups equal weight as an explicitly different metric. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. M scores **70%**; N scores about 81.7%.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: M scores **70%**; N scores about 81.7%. This averages group rates and is not the original 16-card agreement rate.

Repair a missing building block

Review Missing and Unbalanced Examples (ai-1/22). Goal: Name the target collection, selection list and cases actually assessed. Short practice: Would testing only dim plain cards close every current gap?

Point to the target groups.. Return to this week and explain what you changed.

Review Recommendations (ai-1/20). Goal: Apply eligibility requirements before ranking candidates by preferences. Short practice: Why does the recommendation change without changing anyone’s preferences?.

Return to this week and explain what you changed.

Review **Prediction** Is Not Fact (ai-1/17). Goal: Separate a recorded **prediction** from the **observation** used to check it. Short practice: Which field changes the **observation** label, and which field changes the model **prediction**?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

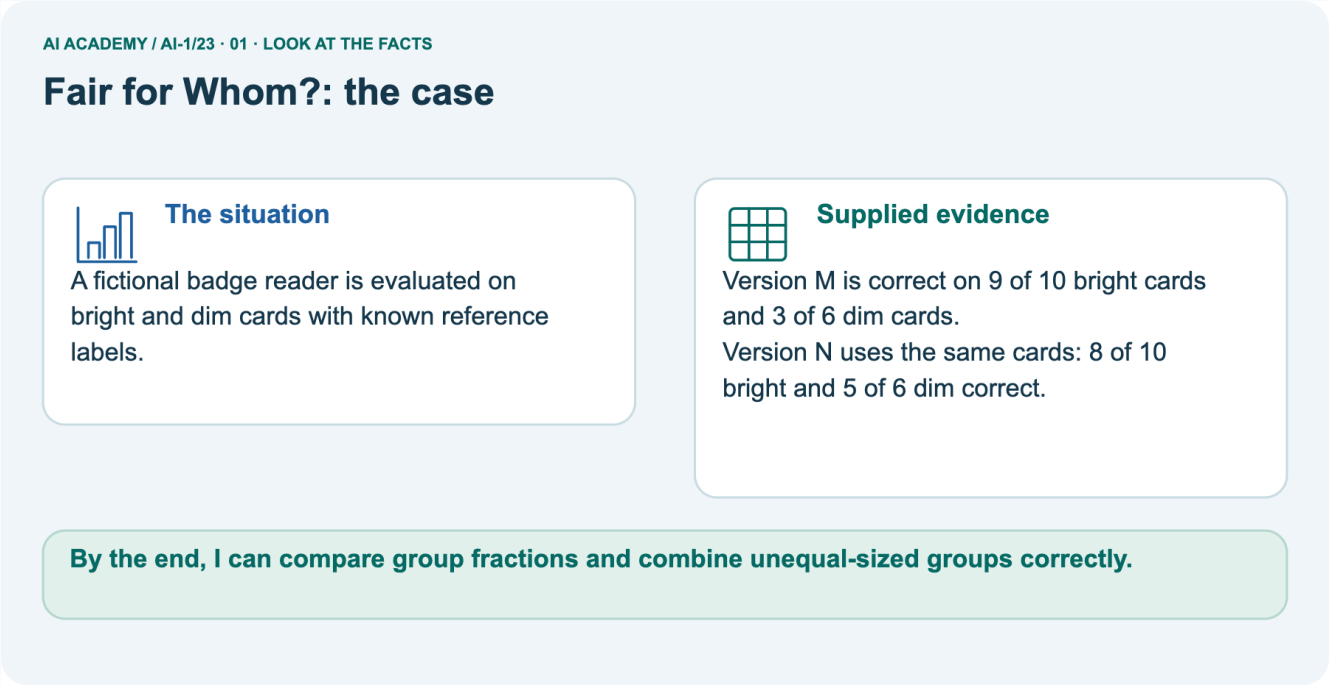
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Fair for Whom? graphic using its labels, deciding evidence and relevant limitation.

01. Fair for Whom?: the case



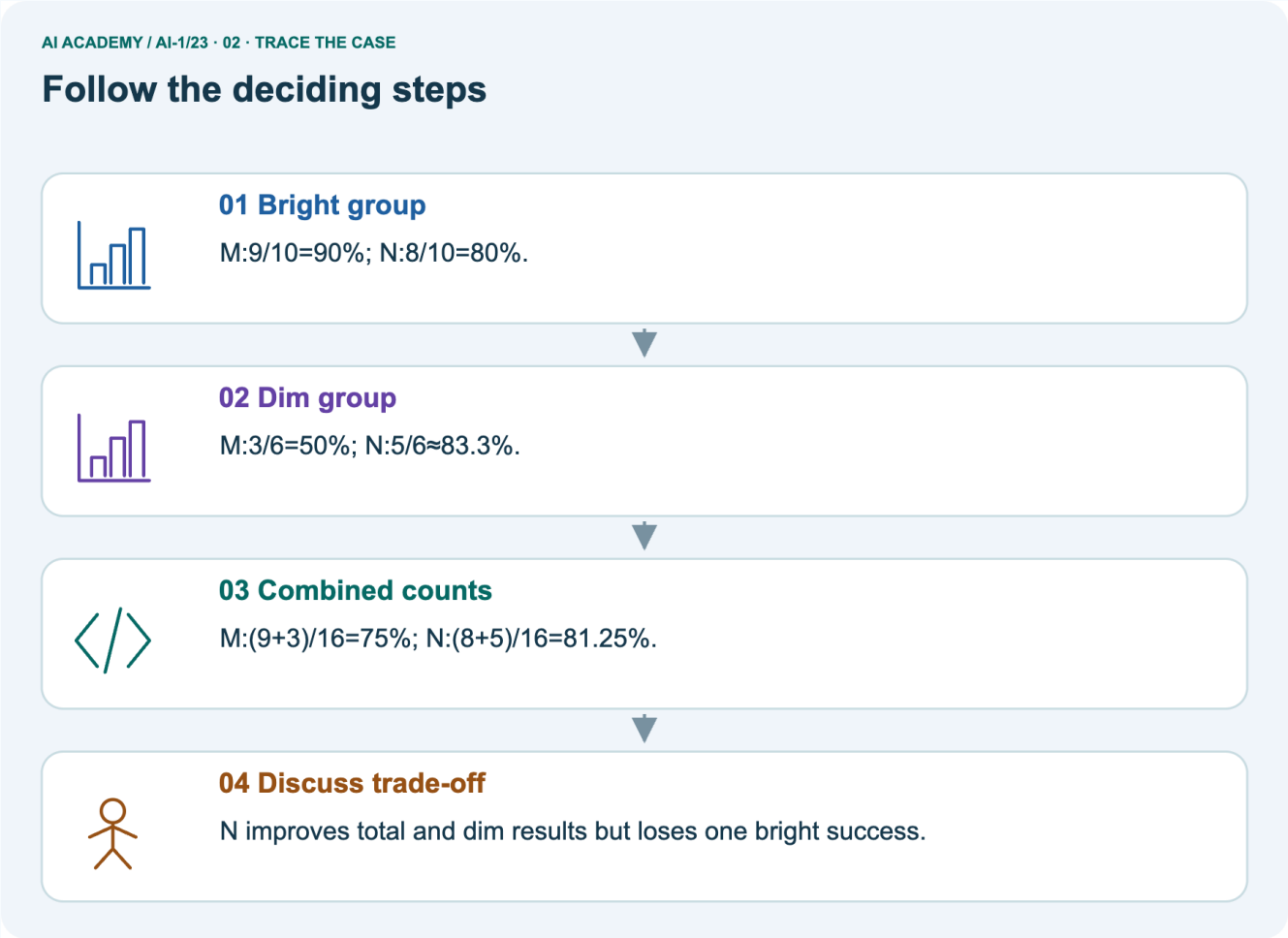
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional badge reader is evaluated on bright and dim cards with known reference labels. Supplied facts: Version M is correct on 9 of 10 bright cards and 3 of 6 dim cards.

Version N uses the same cards: 8 of 10 bright and 5 of 6 dim correct.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

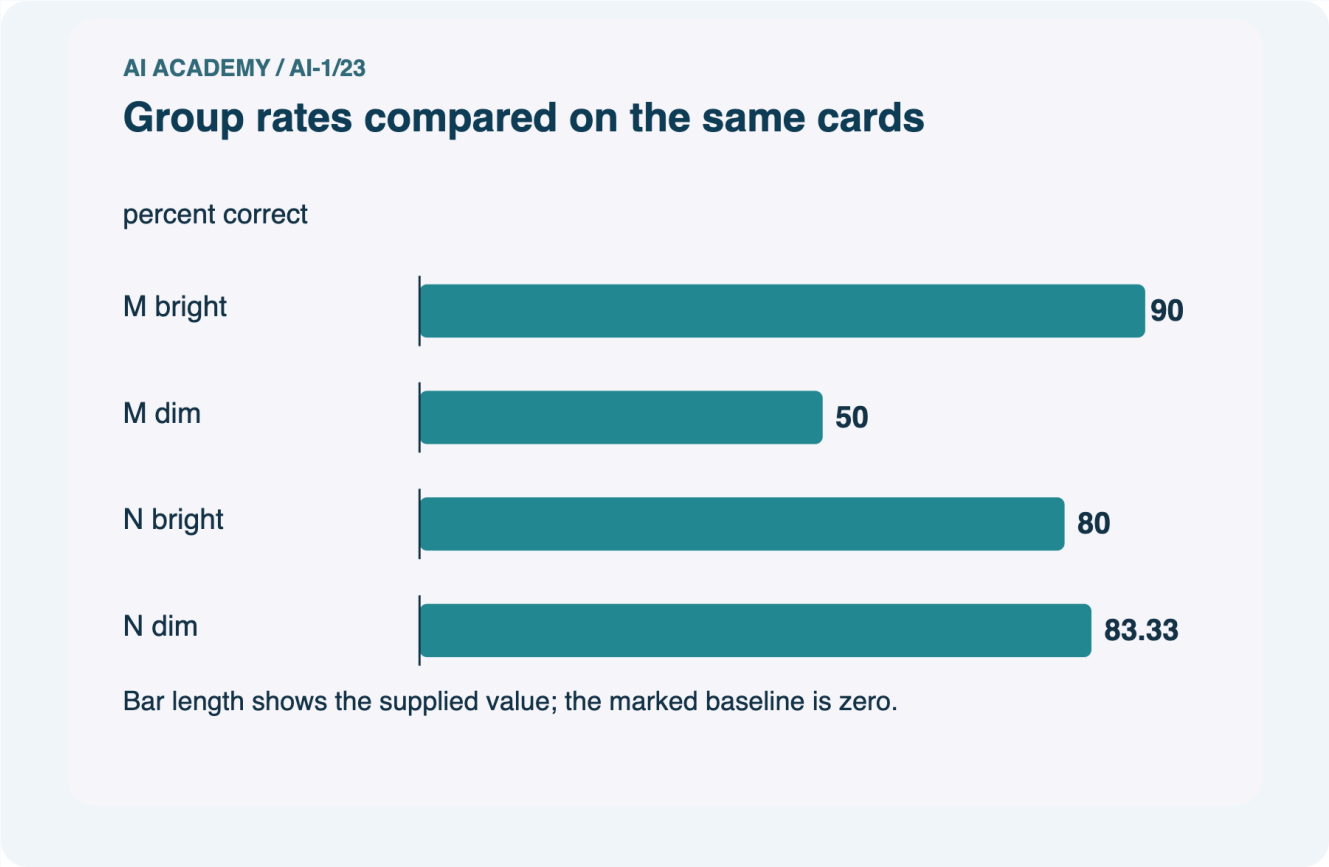


Follow the labelled picture with the matching explanation. Open the vector diagram

Bright group: M:9/10=90%; N:8/10=**80%**. → Dim group: M:3/6=50%; N:5/6≈83.3%. → Combined counts: M:(9+3)/16=75%; N:(8+5)/16=81.25%. → Discuss trade-off: N improves total and dim results but loses one bright success.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Group rates compared on the same cards

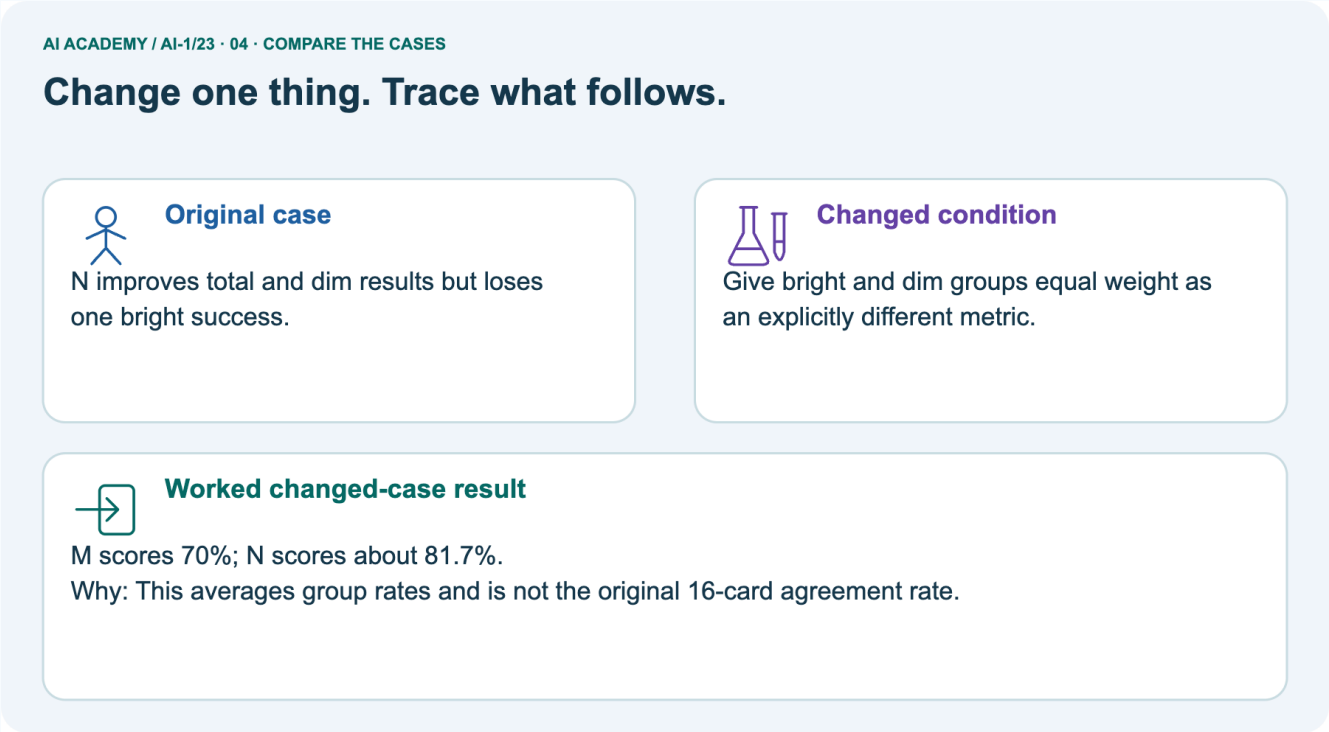


Follow the labelled picture with the matching explanation. Open the vector diagram

Group rates compared on the same cards. Units: percent correct. M bright: 90. M dim: 50. N bright: 80. N dim: 83.33333333333333.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



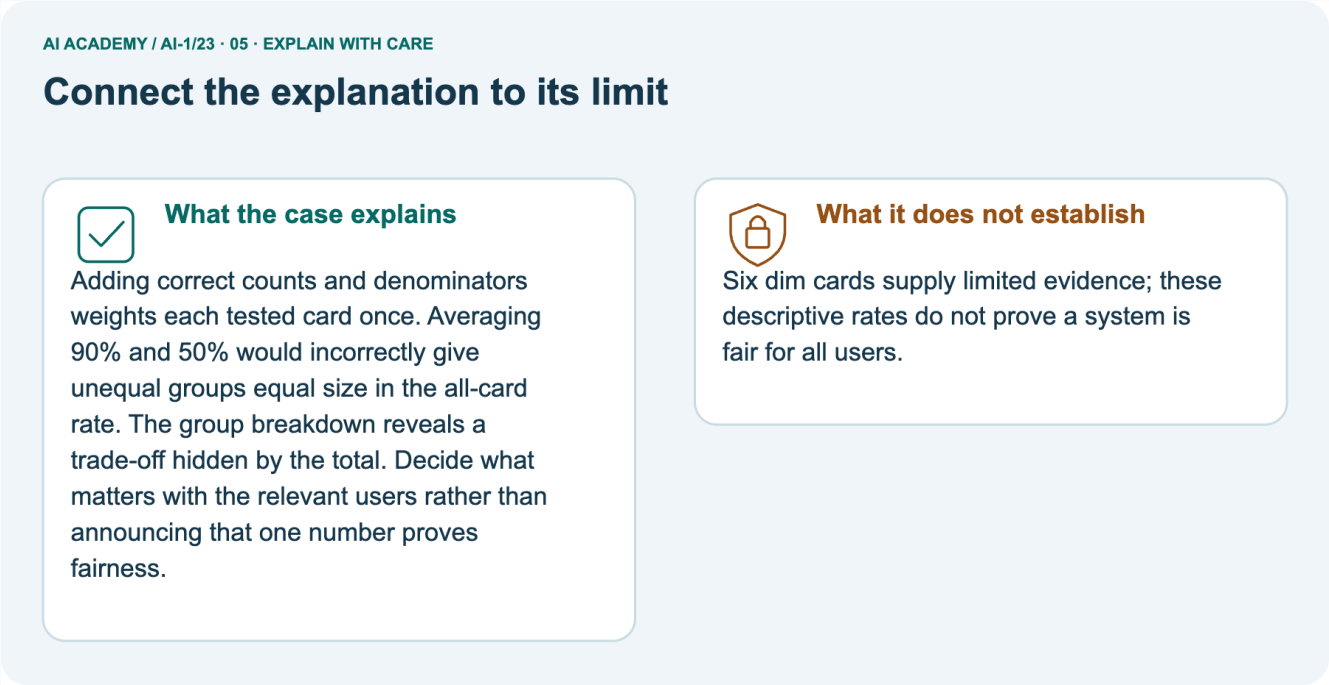
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: N improves total and dim results but loses one bright success. Changed condition: Give bright and dim groups equal weight as an explicitly different metric. Expected result: M scores **70%**; N scores about 81.7%.

Reason: This averages group rates and is not the original 16-card agreement rate.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



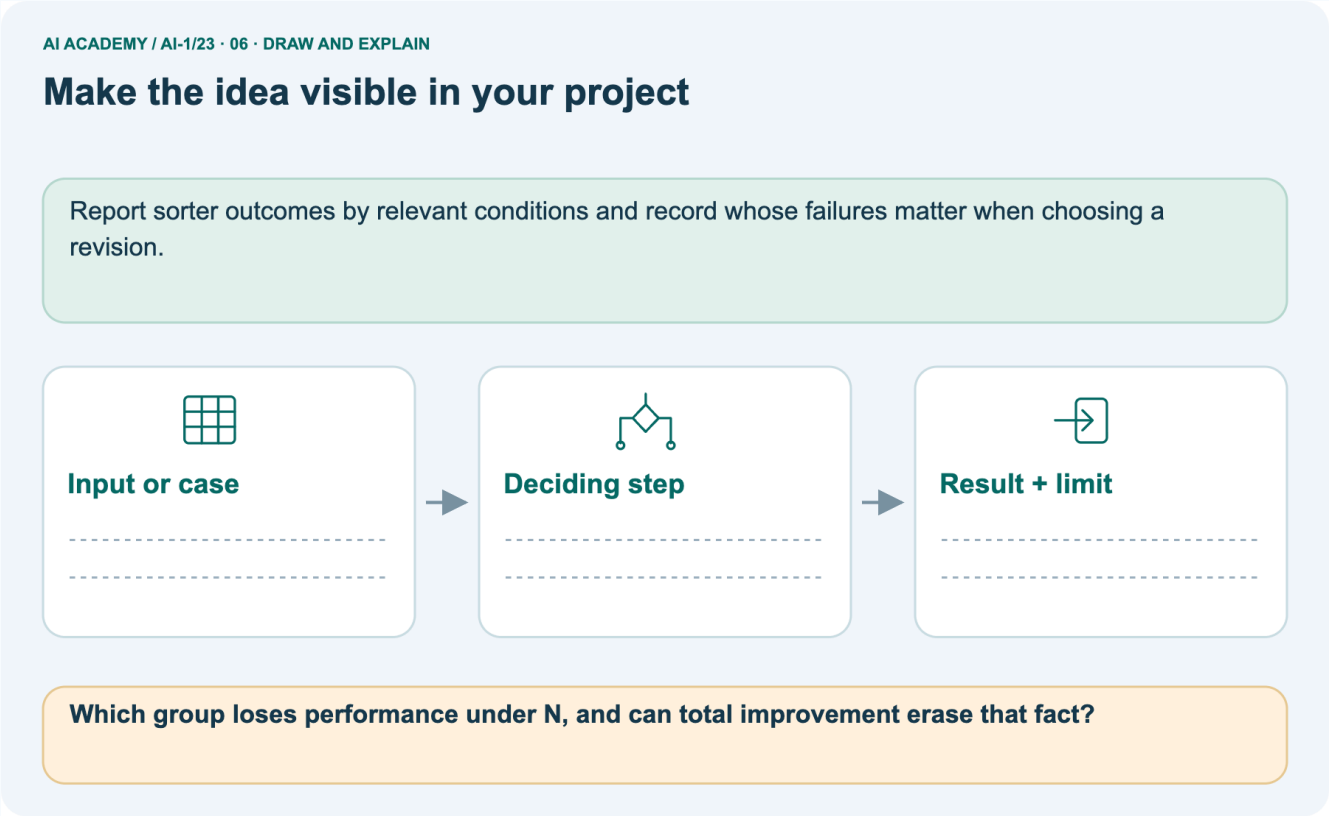
Follow the labelled picture with the matching explanation. Open the vector diagram

Adding correct counts and **denominator**s weights each tested card once. Averaging 90% and 50% would incorrectly give unequal groups equal size in the all-card rate. The group breakdown reveals a trade-off hidden by the total.

Decide what matters with the relevant users rather than announcing that one number proves **fairness**. Limit: Six dim cards supply limited **evidence**; these descriptive rates do not prove a system is fair for all users.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Report sorter outcomes by relevant conditions and record whose failures matter when choosing a revision. Draw your **input**, deciding step, result and limit.

Supported question: Which group loses performance under N, and can total improvement erase that fact?

Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.